Acoustic Vector-reflectivity inversion using first-order parametrization

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Abstract

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Acknowledgements

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A Derivation of the adjoint-state equations and derivative kernel

This appendix shows the calculations of the gradient in greater detail using the adjoint state method based on the works by Plessix [1], Fichtner et al. [2], Cea [3] . The objective is to calculate the functional derivative of the following cost function:

$$J(c, \mathbf{R}, p_s, \mathbf{v}_s) = \frac{1}{2N_sN_rT}\sum_{s=1}^{N_s}\sum_{r=1}^{N_r}\int_0^T \left(S_{s,r}p_s(\mathbf{x},t) - d_{s,r}(t)\right)^2 \, dt, \tag{1}$$

where $c(\mathbf{x})$ is the velocity model, $\mathbf{R} = \{R_1(\mathbf{x}), R_2(\mathbf{x})\}$ is the vector-reflectivity model. Moreover, T is the recording time, $\mathbf{x} = \{x_1, x_2\} \in \Omega$ are the spatial coordinates. Additionally, the state variable $p_s(\mathbf{x}, t)$ is the pressure wavefield, and particle momentum $\mathbf{v}_s = \{v_{s,1}, v_{s,2}\}$, where $s \in \{1, \cdots, N_s\}$ is the shot index and $r \in \{1, \cdots, N_r\}$ is the receiver index. $S_{s,r}$ is a sampling operator that samples the p_s at a specific receiver position r and can be different for every shot index s. The definition of the sampling operator used in this paper is:

$$S_{s,r}p_s = \int_\Omega p_s(\mathbf{x}, t)\, \delta(\mathbf{x} - \mathbf{x}_{s,r})\, d\mathbf{x}, \tag{2}$$

where $\mathbf{x}_r$ is the position of the receiver.

Both velocity (c) and vector-reflectivity ($\mathbf{R}$) are related to the state variables p_s and $v_{s,i}$ by the system of partial diffential equations derived in Soares and Sacchi [4] in form of functional mappings:

$$\frac{\partial v_{s,i}}{\partial t} + \frac{\partial p_s}{\partial x_i} = 0 \tag{3}$$

$$\frac{\partial p_s}{\partial t} + \sum_{i=1}^{2} c\frac{\partial c}{\partial x_i}\, v_{s,i} - 2c^2R_i + c^2\frac{\partial v_{s,i}}{\partial x_i} - f = 0, \tag{4}$$

where if F_1 and F_2 are equal to zero, then the p_s and $v_{s,i}$ are actual physical solutions to this system. In order to make the solution unique, let us use these boundary and initial conditions

$$p_s(\mathbf{x}, 0) = 0, \tag{5}$$

$$v_{s,i}(\mathbf{x}, 0) = 0, \tag{6}$$

$$p_s(\mathbf{x}, t) = 0 \quad \text{at } \mathbf{x} \in \partial\Omega, \tag{7}$$

$$v_{s,i}(\mathbf{x}, t) = 0 \quad \text{at } \mathbf{x} \in \partial\Omega. \tag{8}$$

All of these equations introduce a new problem, how to calculate the derivative of the cost function in Eq. 1 when we have an implicit relationship between the state variables and parameters in the form of a PDE? This is solved by using the adjoint state method.

The first step in the adjoint state method is to assemble the augmented Lagrangian that turns the system of PDEs and boundary

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and initial conditions into regularization terms:

$$
\begin{aligned}
&\mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_{\mathbf{s}}) = J(c, \mathbf{R}, p_s, \mathbf{v}_s) \\
&\quad - \sum_{s=1}^{N_s} \sum_{i=1}^{2} \int_\Omega \int_0^T \xi_{s,i} \left(\frac{\partial v_{s,i}}{\partial t} + \frac{\partial p_s}{\partial x_i} \right) dt\, d\mathbf{x} \\
&\quad - \sum_{s=1}^{N_s} \int_\Omega \int_0^T \eta_s \left(\frac{\partial p_s}{\partial t} + \sum_{i=1}^{N_s} c \frac{\partial c}{\partial x_i} v_{s,i} - 2c^2 R_i + c^2 \frac{\partial v_{s,i}}{\partial x_i} - f \right) dt\, d\mathbf{x}, \\
&\quad - \sum_{s=1}^{N_s} \int_{\partial\Omega} \int_0^T \gamma_s(\mathbf{x}, t) p_s(\mathbf{x}, t)\, dt\, d\mathbf{x} - \sum_{s=1}^{N_s} \int_0^T \mu_s(\mathbf{x}) p_s(\mathbf{x}, 0)\, dt \\
&\quad - \sum_{s=1}^{N_s} \int_{\partial\Omega} \int_0^T \zeta_{s,i}(\mathbf{x}, t) v_{s,i}(\mathbf{x}, t)\, dt\, d\mathbf{x} - \sum_{s=1}^{N_s} \int_0^T \lambda_{s,i}(\mathbf{x}) v_{s,i}(\mathbf{x}, 0)\, dt
\end{aligned} \tag{9}
$$

with $\xi_{s,i}$, η_s being the adjoint state variables for the PDEs, and γ_s, μ_s, $\zeta_{s,i}$, and $\lambda_{s,i}$ are the adjoint state variables for the initial and boundary conditions.

The next step is to calculate the directional functional derivative of the Lagrangian, which we use the definition from Zeidler [5, p. 228]:

$$
\frac{dF}{du} h = \lim_{\epsilon \to 0} \frac{F(u + \epsilon h) - F(u)}{\epsilon} = \left[\frac{d}{d\epsilon} F(u + \epsilon h) \right]_{\epsilon=0}, \tag{10}
$$

where $\frac{dF}{du}$ is a linear operator. In order to facilitate the mathematical constraints and avoid lack of rigor, we will go ahead and assume that $F(u)$ will be always continuous, which means that $\frac{dF}{du}$ will remain the same in all directions and that it exists. Let us also define a "regular" and "partial" variants for the multivariate case. With the regular we assume that all the variables are dependent on each other, which means that the directional functional derivative becomes:

$$
\begin{aligned}
&\frac{d}{dm} F(m, u_1, \cdots, u_N)\, h = \\
&\qquad \left[\frac{d}{d\epsilon} F(m + \epsilon h, u_1(m + \epsilon h), \cdots, u_N(m + \epsilon h)) \right]_{\epsilon=0},
\end{aligned} \tag{11}
$$

while in the partial case one only considers that specific variable:

$$
\frac{\partial}{\partial m} F(m, u_1, \cdots, u_N)\, h = \left[\frac{d}{dt} F(m + \epsilon h, u_1, \cdots, u_N) \right]_{\epsilon=0}. \tag{12}
$$

These definitions are important because the regular derivative of the cost function with relation to the model is equal to the partial derivative of the lagrangian in Eq. 9, so long that the partial derivatives with relation to all the other variables is set to zero. This can be expressed as

$$
\begin{aligned}
&\frac{d}{d\mathbf{R}} J(c, \mathbf{R}, p_s, \mathbf{v}_s) = \frac{\partial}{\partial \mathbf{R}} \mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_{\mathbf{s}}) \iff \\
&\qquad \frac{\partial}{\partial \alpha} \mathcal{L} = 0 \quad \text{for } \alpha \in \{p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_{\mathbf{s}}\}.
\end{aligned} \tag{13}
$$

In this case, one would also have to set $\frac{d}{dc}\mathcal{L} = 0$, but that does not lead to an equation that needs solving on top of the adjoint state equations given by $\frac{\partial}{\partial \alpha}\mathcal{L} = 0$, and this paper assumes that $\mathbf{R}$ and c are not related.

The adjoint state equations are calculated by equaling the derivatives the augmented Lagrangian (Eq. 9) with relation to the state variables p_s and $v_{s,i}$ to zero. The derivative with relation to p_s is calculated as follows using the definition:

$$
\begin{aligned}
&\frac{\partial}{\partial p_s} \mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_{\mathbf{s}})\, h_s = \frac{\partial}{\partial p_s} J(c, \mathbf{R}, p_s, \mathbf{v}_s)\, h_s \\
&\quad - \sum_{s=1}^{N_s} \sum_{i=1}^{2} \int_\Omega \int_0^T \xi_{s,i} \left(\frac{\partial}{\partial x_i} h_s \right) dt\, d\mathbf{x} \\
&\quad - \sum_{s=1}^{N_s} \int_\Omega \int_0^T \eta_s \left(\frac{\partial}{\partial t} h_s \right) dt\, d\mathbf{x} \\
&\quad - \sum_{s=1}^{N_s} \int_{\partial\Omega} \int_0^T \gamma_s(\mathbf{x}, t) h_s(\mathbf{x}, t)\, dt\, d\mathbf{x} - \sum_{s=1}^{N_s} \int_0^T \mu_s(\mathbf{x}) h_s(\mathbf{x}, 0)\, dt.
\end{aligned} \tag{14}
$$

The equation above is a derivative with relation to all possible p_s, for every shot index (p_1, p_2, p_3), whereas we only need the derivative for a specific one for the sought after adjoint state equation. Hence, it is strategic here to use the Kronecker delta to enforce that inside the definition of the test function h_s like so:

$$
h_s(\mathbf{x}, t) = \delta_{s,\bar{s}}\, \psi(\mathbf{x}, t), \tag{15}
$$

which when substituted back into Eq. 14 results into

$$
\begin{aligned}
&\frac{\partial}{\partial p_{\bar{s}}} \mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_{\mathbf{s}})\, (\delta_{s,\bar{s}}\, \psi(\mathbf{x}, t)) = \\
&\qquad \frac{\partial}{\partial p_{\bar{s}}} J(c, \mathbf{R}, p_s, \mathbf{v}_s)\, (\delta_{s,\bar{s}}\, \psi(\mathbf{x}, t)) \\
&\quad - \sum_{i=1}^{2} \int_\Omega \int_0^T \xi_{\bar{s},i} \left(\frac{\partial}{\partial x_i} \psi \right) dt\, d\mathbf{x} - \int_\Omega \int_0^T \eta_{\bar{s}} \left(\frac{\partial}{\partial t} \psi \right) dt\, d\mathbf{x} \\
&\quad - \int_{\partial\Omega} \int_0^T \gamma_{\bar{s}}(\mathbf{x}, t) \psi(\mathbf{x}, t)\, dt\, d\mathbf{x} - \int_0^T \mu_{\bar{s}}(\mathbf{x}) \psi(\mathbf{x}, 0)\, dt,
\end{aligned} \tag{16}
$$

where now the derivative is in relation to a specific shot index $\bar{s}$. The next step is to apply integration by parts in the relevant integrals to move every differential operator away from the test function ψ:

$$
\begin{aligned}
&\frac{\partial}{\partial p_{\bar{s}}} \mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_{\mathbf{s}})\, (\delta_{s,\bar{s}}\, \psi(\mathbf{x}, t)) = \\
&\qquad \frac{\partial}{\partial p_{\bar{s}}} J(c, \mathbf{R}, p_s, \mathbf{v}_s)\, (\delta_{s,\bar{s}}\, \psi(\mathbf{x}, t)) \\
&\quad - \sum_{i=1}^{2} \int_{\partial\Omega} \int_0^T \xi_{\bar{s},i}\, \psi\, dt\, d\mathbf{x} + \sum_{i=1}^{2} \int_\Omega \int_0^T \left(\frac{\partial}{\partial x_i} \xi_{\bar{s},i} \right) \psi\, dt\, d\mathbf{x} \\
&\quad - \int_\Omega \eta_{\bar{s}}(\mathbf{x}, T) \psi(\mathbf{x}, T)\, d\mathbf{x} \\
&\quad + \int_\Omega \eta_{\bar{s}}(\mathbf{x}, 0) \psi(\mathbf{x}, 0)\, d\mathbf{x} + \int_\Omega \int_0^T \left(\frac{\partial}{\partial t} \eta_{\bar{s}} \right) \psi\, dt\, d\mathbf{x} \\
&\quad - \int_{\partial\Omega} \int_0^T \gamma_{\bar{s}}(\mathbf{x}, t) \psi(\mathbf{x}, t)\, dt\, d\mathbf{x} - \int_0^T \mu_{\bar{s}}(\mathbf{x}) \psi(\mathbf{x}, 0)\, dt,
\end{aligned} \tag{17}
$$

and here we can define the adjoint variables for the boundary conditions to be $\gamma_{\bar{s}}(\mathbf{x}, t) = -\sum_{i=1}^{2} \xi_{\bar{s},i}(\mathbf{x}, t)$, and $\mu_{\bar{s}}(\mathbf{x}) = \eta_{\bar{s}}(\mathbf{x}, 0)$. That way, the boundary and initial terms cancel if we also set $\eta_s(\mathbf{x}, T) = 0$.

$$\frac{\partial}{\partial p_{\bar{s}}}\mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_\mathbf{s})\ (\delta_{s,\bar{s}}\,\psi(\mathbf{x}, t)) = \frac{\partial}{\partial p_{\bar{s}}}J(c, \mathbf{R}, p_s, \mathbf{v}_s)\ (\delta_{s,\bar{s}}\,\psi(\mathbf{x}, t)) + \sum_{i=1}^{2}\int_\Omega\int_0^T\left(\frac{\partial}{\partial x_i}\xi_{\bar{s},i}\right)\psi\,dt\,d\mathbf{x} + \int_\Omega\int_0^T\left(\frac{\partial}{\partial t}\eta_{\bar{s}}\right)\psi\,dt\,d\mathbf{x} \tag{18}$$

which leave us with the only with the derivative of the cost function. This derivative is calculated as follows:

$$\frac{\partial}{\partial p_{\bar{s}}}J(c, \mathbf{R}, p_s, \mathbf{v}_s)\ (\delta_{s,\bar{s}}\,\psi(\mathbf{x}, t)) = \left[\frac{d}{d\epsilon}\frac{1}{\beta}\sum_{s=1}^{N_s}\sum_{r=1}^{N_r}\int_0^T\left(S_{s,r}\left(p_s(\mathbf{x}, t) + \epsilon\,\delta_{s,\bar{s}}\,\psi(\mathbf{x}, t)\right) - d_{s,r}(t)\right)^2 dt\right]_{\epsilon=0}, \tag{19}$$

with $\beta = \frac{1}{2N_sN_rT}$. Given this is a derivative to $\epsilon \in \mathbb{R}$, it can go inside the sums and integrals, leading to:

$$\frac{\partial}{\partial p_{\bar{s}}}J(c, \mathbf{R}, p_s, \mathbf{v}_s)\ (\delta_{s,\bar{s}}\,\psi(\mathbf{x}, t)) = \frac{1}{\beta}\sum_{r=1}^{N_r}\int_0^T 2\Big(S_{\bar{s},r}\left(p_{\bar{s}}(\mathbf{x}, t)\right) - d_{\bar{s},r}(t)\Big)S_{\bar{s},r}\,\psi(\mathbf{x}, t)\,dt. \tag{20}$$

The objective here is to put Eq. 20 with the same space-time integrals of the other terms in Eq. 18, which will require playing around with the sampling operator definition:

$$\begin{aligned}
&\frac{\partial}{\partial p_{\bar{s}}}J(c, \mathbf{R}, p_s, \mathbf{v}_s)\ (\delta_{s,\bar{s}}\,\psi(\mathbf{x}, t))\\
&= \frac{1}{\beta}\sum_{r=1}^{N_r}\int_0^T 2\Big(S_{\bar{s},r}\left(p_{\bar{s}}(\mathbf{x}', t)\right) - d_{\bar{s},r}(t)\Big)\int_\Omega\psi(\mathbf{x}, t)\delta(\mathbf{x}-\mathbf{x}_{s,r})\,d\mathbf{x}\,dt\\
&= \frac{2}{\beta}\sum_{r=1}^{N_r}\int_0^T\int_\Omega\Big(S_{\bar{s},r}\left(p_{\bar{s}}(\mathbf{x}', t)\right) - d_{\bar{s},r}(t)\Big)\delta(\mathbf{x}-\mathbf{x}_{s,r})\,\psi(\mathbf{x}, t)\,d\mathbf{x}\,dt\\
&= \frac{2}{\beta}\sum_{r=1}^{N_r}\int_0^T\int_\Omega S^\dagger_{\bar{s},r}\Big(S_{\bar{s},r}\left(p_{\bar{s}}(\mathbf{x}', t)\right) - d_{\bar{s},r}(t)\Big)\psi(\mathbf{x}, t)\,d\mathbf{x}\,dt,
\end{aligned} \tag{21}$$

where $S^\dagger_{\bar{s},r}$ is the adjoint of the sampling operator, which just means placing a value at the position of the receiver. Substituting the equation above (Eq. 20) into the expression of the derivative of the lagrangian with relation to the pressure (Eq. 18) while equaling to zero to satisfy the adjoint state method conditions results into:

$$\frac{\partial}{\partial p_{\bar{s}}}\mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_\mathbf{s})\ (\delta_{s,\bar{s}}\,\psi(\mathbf{x}, t)) = \frac{2}{\beta}\sum_{r=1}^{N_r}\int_0^T\int_\Omega S^\dagger_{\bar{s},r}\Big(S_{\bar{s},r}\left(p_{\bar{s}}(\mathbf{x}', t)\right) - d_{\bar{s},r}(t)\Big)\psi(\mathbf{x}, t)\,d\mathbf{x}\,dt + \sum_{i=1}^{2}\int_\Omega\int_0^T\left(\frac{\partial}{\partial x_i}\xi_{\bar{s},i}\right)\psi\,dt\,d\mathbf{x} + \int_\Omega\int_0^T\left(\frac{\partial}{\partial t}\eta_{\bar{s}}\right)\psi\,dt\,d\mathbf{x} = 0, \tag{22}$$

which is a weak/integral formulation of the following PDE:

$$\frac{\partial\eta_s}{\partial t} + \sum_{i=1}^{2}\frac{\partial\xi_{s,i}}{\partial x_i} + \frac{2}{\beta}\sum_{r=1}^{N_r}S^\dagger_{s,r}\Big(S_{s,r}\left(p_s(\mathbf{x}, t)\right) - d_{s,r}(t)\Big) = 0 \tag{23}$$

where the $\bar{s}$ were replaced with s to simplify notation, given now it is clear that the equation is the same for any shot index. Calculating the derivative of the lagrangian in Eq. 9 with relation to the particle momentum $v_{s,i}$, for a specific shot index, will result into the other adjoint state equation:

$$\frac{\partial}{\partial t}\xi_{s,i} + \frac{\partial}{\partial x_i}\left(c^2\eta\right) + \eta\left(2c^2R_i - c\frac{\partial}{\partial x_i}c\right) = 0, \tag{24}$$

which together with Eq. 23 compose the adjoint system of equations to be solved for $\xi_{s,i}$ and η_s.

The last step is to calculate the derivative of the lagrangian (Eq. 9) with relation to the reflectivity, to a specific component $\bar{i}$, in a similar fashion to the other derivatives:

$$\begin{aligned}
&\frac{\partial}{\partial R_{\bar{i}}}\mathcal{L}(c, \mathbf{R}, p_s, \mathbf{v}_s, \xi_s, \eta_s, \gamma_s, \mu_s, \zeta_s, \lambda_\mathbf{s})\left(\delta_{i,\bar{i}}\,\phi\right)\\
&= 2\int_0^T\int_\Omega c^2(\mathbf{x})\,\eta(\mathbf{x}, t)\,v_{s,\bar{i}}(\mathbf{x}, t)\,\phi(\mathbf{x})\,d\mathbf{x}\,dt\\
&= \left\langle 2\int_0^T c^2(\mathbf{x})\,\eta(\mathbf{x}, t)\,v_{s,\bar{i}}(\mathbf{x}, t)\,dt,\ \phi(\mathbf{x})\right\rangle_\Omega,
\end{aligned} \tag{25}$$

where $\phi(\mathbf{x})$ is just a test function. The integral over space in this directional functional derivative is a inner product of the functional derivative kernel (Fréchet, non-directional) with the test function, which collapses the space variable and makes the resulting test derivative be a single scalar for the whole space Ω. When discretized, this non-directional derivative becomes a jacobian, while the directional version becomes the dot product of the jacobian with a test vector. Strategically here, a simple way to obtain the derivative one can simply choose the test function to be a Dirac delta at another point in space, and use the its sampling property:

$$\phi(\mathbf{x}) = \delta(\mathbf{x} - \mathbf{x}'), \tag{26}$$

which when applied to the Eq. 25 leads to the sought after derivative of the cost function:

$$\frac{d}{dR_i}J = 2\int_0^T c^2(\mathbf{x})\,\eta(\mathbf{x}, t)\,v_{s,i}(\mathbf{x}, t)\,dt, \tag{27}$$

where again it is clear that the expression for the derivative is the same for different components (i), hence the equation above has i instead of $\bar{i}$. The equation above (Eq. 27), when discretized is the gradient of the cost function with relation to the vector-reflectivity.